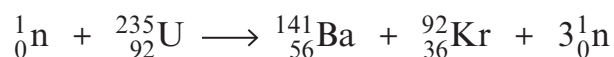


SECTION AAnswer **all** questions.

1. A fission reaction of uranium is:



Masses:

$${}_{92}^{235}\text{U} = 235.043\,930\,\text{u}$$

$${}_{56}^{141}\text{Ba} = 140.914\,412\,\text{u}$$

$${}_{36}^{92}\text{Kr} = 91.926\,156\,\text{u}$$

$$m_{\text{neutron}} = 1.008\,665\,\text{u}$$

$$1\,\text{u} = 931\,\text{MeV}$$

- (a) Calculate the energy released in this reaction giving your answer in MeV. [4]

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- (b) If 1.2 kg of
- ${}_{92}^{235}\text{U}$
- undergoes this reaction, show that the total energy released is approximately
- $8.5 \times 10^{13}\,\text{J}$
- . [3]

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- (c) An engineer at a nuclear power station claims that over two billion 11 W light bulbs could be powered for 1 hour by the energy released in part (b). Discuss whether or not her claim is correct. [One billion = 1×10^9] [3]

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